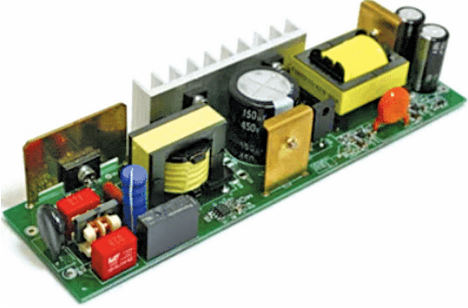


Interesting Reference Designs Of 100W LED DRIVER

Title of Reference Design	100W LED Commercial Power Supply	100W Flyback LED Ballast
Image		
Key Attractions	The reference design gives a steady output, uses less energy, and handles heat and noise well. It connects directly to a heat sink, which makes it easy to build and keeps it cool. Its small size and 90% efficiency make it a good fit for many lighting setups.	The reference design works with different input voltages and supports 3-way dimming. It saves energy, reduces heat, and helps lower the overall cost. Built-in safety features protect the system from damage, and the design supports smooth lighting with low signal noise.
Highlights	This Toshiba reference design outlines a 100W power supply for commercial lighting systems, focusing on electrical performance, heat handling, and EMI control. Commercial setups use separate power units, making space and thermal management easier. The design uses a through-hole insulated package for direct heat sink mounting without extra insulation. It includes two stages: a power factor correction (PFC) circuit that raises AC input to about 390V DC, and a flyback converter that lowers it to 110V DC at 1.04A. MOSFETs are selected for voltage and thermal handling. For protection and EMI control, the design uses a ceramic varistor, X capacitors, a common mode choke, and discharge resistors—adjustable based on PCB layout and case. A bridge diode handles rectification and inrush currents. At 90% efficiency under 230V input and full load, the design reduces energy loss and heat. With a size of 180mm×57mm×40mm, it supports a range of lighting uses. It also offers guidance on part selection, noise control, and heat management for LED systems.	RDK-801 is a 100W reference design from Power Integrations for an LED ballast with 3-way dimming. It delivers constant current for LED strings from 36V to 48V and supports output currents from 0mA to 2080mA. The PFC stage integrates a CCM controller, gate driver, and 600V MOSFET in one package, removing the need for external current-sense resistors and adjusting switching frequency based on input and load. The flyback stage includes the primary switch, control logic, and synchronous rectification, and uses FluxLink to transmit signals across the isolation barrier without an optocoupler. The input section includes a fuse, varistor, NTC thermistor, EMI filter, and bridge rectifier to support low THD and high power factor. Integrated ICs reduce the need for separate components, lowering PCB size and improving reliability. The PFC stage maintains efficiency across all loads, meeting power factor and THD targets without extra circuitry. Built-in protections such as over-voltage and over-current simplify compliance and improve fault handling. The design reduces cost and development time for manufacturers and lowers energy use and maintenance for end users.
Applications	The reference design is suitable for a wide range of commercial lighting applications.	The reference design is suitable for global applications like street lighting and commercial lighting.
OEM Brand	Toshiba	Power Integration
URL	https://www.electronicsforu.com/electronics-projects/reference-design-for-100w-led-commercial-power-supply	https://www.electronicsforu.com/electronics-projects/100-w-flyback-led-ballast-reference-design